
EyeGaze Inference

Real-Time Gaze Estimation with GazeDINO

DINOv2 ViT-B/14 + MediaPipe FaceMesh + Polynomial Calibration

Best model: 3.24° mean angular error

Emanuele Di Luzio

Model Architecture — GazeDINO

Input
(B, 3, 224, 224)

DINOv2 ViT-B/14
86M params
Self-supervised on 142M imgs

CLS Token [768]

MLP Head
LayerNorm(768)
768 → 512 (GELU, drop 0.30)
512 → 128 (GELU, drop 0.15)
128 → 2

(yaw, pitch) radians

Emanuele Di Luzio

Training Details

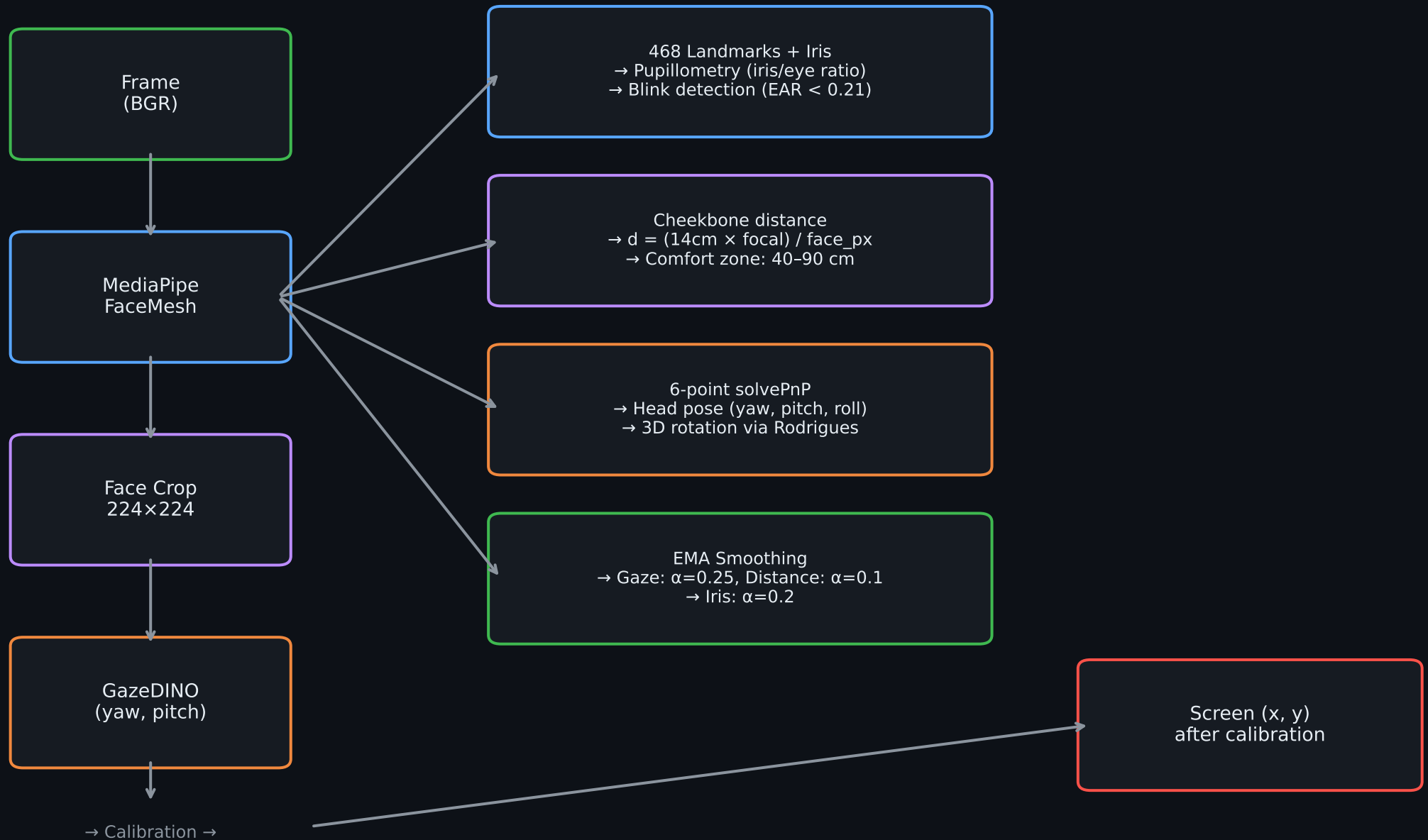
- Dataset: GazeGene (56 subjects, 9 cameras, ~90K samples)
- Loss: Angular error (mean angle between 3D gaze vectors)
- Best val error: 3.24°
- Epochs: 50, optimizer: AdamW
- Backbone frozen for first epochs, then fine-tuned

Backbone Options

Backbone	Params	Embed
dinov2_vits14	21M	384
dinov2_vitb14	86M	768 ← default
dinov2_vitl14	307M	1024
dinov2_vitg14	1.1B	1536

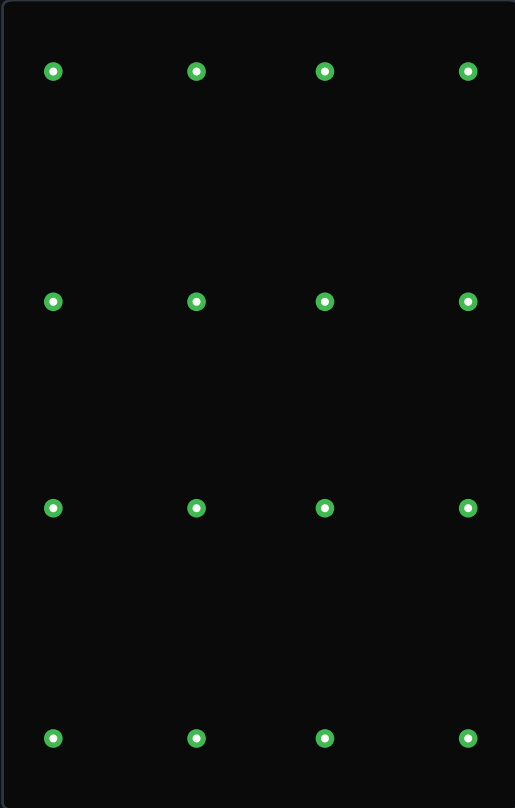
EyeGaze Inference

Inference Pipeline



16-Point Gaze Calibration

4×4 Calibration Grid



How It Works

1. Fullscreen overlay shows 16 dots (4×4 grid)
2. Follow each dot with your eyes (2.8s per point)
3. Samples collected after 35% of dwell time
4. Outlier filtering: remove samples $> 1.5\sigma$ from median
5. Degree-3 polynomial least-squares fit (10 coeff/axis)
6. Min 10 valid points required
7. Saved to calibration.pkl for reuse

Polynomial Features (degree 3)

$[1, y, p, y^2, yp, p^2, y^3, y^2p, yp^2, p^3]$

Two separate fits: features $\rightarrow$ screen_x, features $\rightarrow$ screen_y

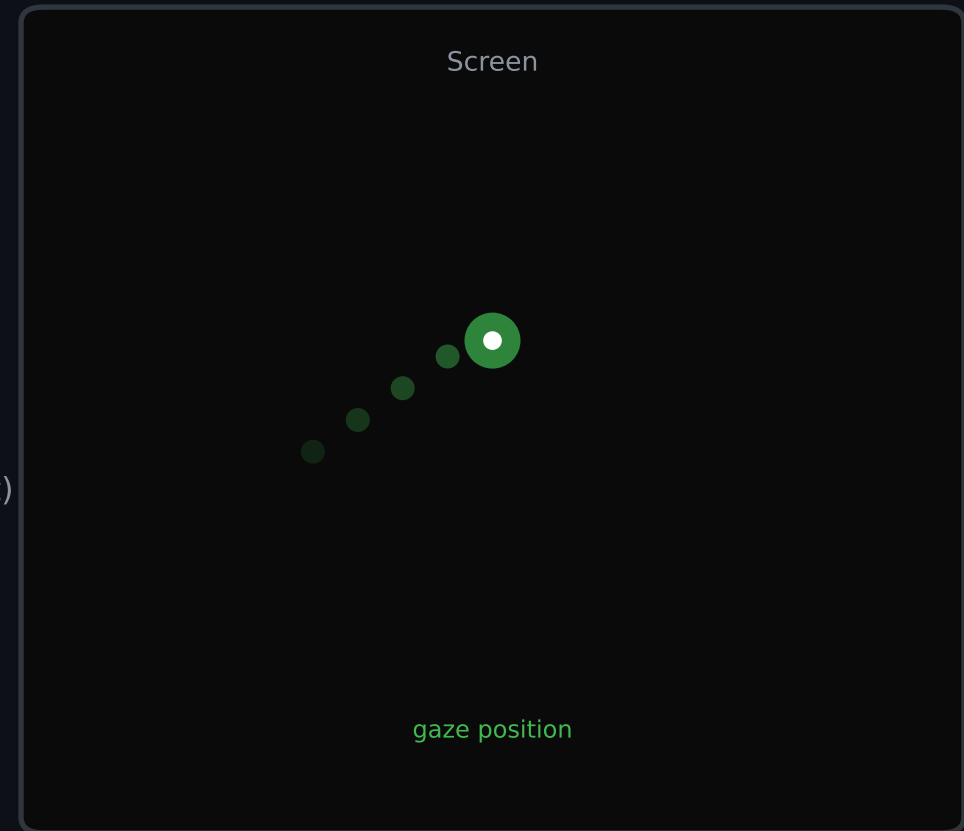
Screen Gaze — Real-Time Overlay

How It Works

1. Calibrate (16 points, ~45s total)
2. Enable 'Screen gaze' toggle in sidebar
3. Translucent green dot appears on screen
4. Follows your gaze in real-time
5. Position smoothed with EMA ($\alpha=0.5$)
6. macOS: transparent background (systemTransparent)
7. Always-on-top overlay window

Pipeline

GazeDINO → (yaw, pitch) → EMA → Poly3 calibrator → (screen_x, screen_y) → overlay dot



GUI & Features

Dark-Mode Interface

- Built with CustomTkinter
- Live camera feed with overlays
- Sidebar: gaze, metrics, calibration, toggles
- Multi-face support (up to 5 faces)
- Stable face ordering via centroid tracking
- Video mode: open files, seek, play/pause

Overlay Toggles

- Face box
- Gaze arrows
- Iris dots
- Screen gaze (post-calibration)

Live Metrics

Yaw / Pitch

Model output → degrees

Distance

Pinhole model, 40–90 cm zone

Iris ratio

Pupillometry proxy

EAR

Blink detection (thresh 0.21)

Head pose

solvePnP → yaw, pitch, roll

Screen XY

After calibration only

Keyboard Shortcuts

- | | |
|-------|-------------|
| Q | Quit |
| C | Calibrate |
| Space | Play/Pause |
| ← → | Switch face |

Summary & Technical Stack

Tech Stack

Key Numbers

PyTorch + TorchVision

DINOv2 backbone, inference

3.24°

Mean angular error

MediaPipe

FaceMesh 468 landmarks + iris

86M

DINOv2 ViT-B/14 parameters

OpenCV

Video capture, drawing, solvePnP

16

Calibration points (4×4)

CustomTkinter

Dark-mode GUI

10

Polynomial coefficients per axis

NumPy

Polynomial fit, EMA, linear algebra

5

Max simultaneous faces

~30 fps

Real-time on Apple Silicon (MPS)

End-to-End Pipeline

Webcam → MediaPipe FaceMesh → Face Crop → GazeDINO (yaw, pitch) → EMA → Poly3 Calibration → Screen (x, y) → Overlay Dot

github.com/emanuelediluzio/eyegaze-inference